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Tutorial 2 - Apache Spark

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1.0 Business Understanding

This section establishes the foundation of the project by outlining the context and motivation behind it. It covers the problem being addressed, the objectives the solutions aim to achieve, and the key stakeholders who stand to benefit from the outcomes.

1.1 Background and Problem Statement

Brazil's federal government makes yearly education data available to the public through its Open Data portal, sourced from the National Basic Education Census (Censo Escolar). This census captures comprehensive information about every basic education institution across the country, spanning structural, geographical, ethnic, and socioeconomic dimensions over an eleven-year period from 2010 to 2020. Each annual file contains roughly 370 attributes and upwards of 300,000 records, with the entire collection amounting to approximately 2.2 GB of raw CSV data.

A significant challenge arises from the nature of this data: educators, policymakers, and researchers struggle to run meaningful analytical queries against large, unstructured flat files. Questions such as "How many students were enrolled in kindergarten in 2020?" or "Which states saw the greatest growth in school enrolment between 2010 and 2020?" cannot be answered efficiently from raw CSVs. These files lack the structural organisation required for fast aggregations, and their sheer volume makes conventional tools like Pandas impractical without severe performance trade-offs.

1.2 Business Objectives

This project aims to address the limitations described above through the following goals:

- Develop a robust, scalable ETL pipeline powered by Apache Spark (PySpark) to ingest and process large volumes of education census data.
- Transform the flat CSV structure into a Star Schema dimensional model and persist it within a PostgreSQL database, enabling efficient analytical queries.
- Support business intelligence reporting and interactive dashboards by connecting a Metabase instance to the restructured database.

- Illustrate the practical benefits of Apache Spark's lazy evaluation and distributed computing model over conventional single-machine approaches when handling big data workloads.

1.3 Stakeholders

Several groups stand to benefit from this pipeline. Government education departments and policymakers rely on enrollment trend data to make informed decisions about resource distribution. Academic researchers studying Brazil's public education landscape require consistent, queryable datasets spanning multiple years. Data engineers need a maintainable and extensible pipeline architecture. Finally, BI analysts depend on the structured output to build dashboards and generate reports that drive strategic insight.

2.0 Data Understanding

This section describes the dataset used in this project, including where it originates, how it is structured, and how its contents are logically organised. It examines the key attributes present in the data and outlines the conceptual relationships between entities prior to transformation.

2.1 Data Source and Description

The dataset originates from Brazil's official Open Data portal, maintained by INEP (Instituto Nacional de Estudos e Pesquisas Educacionais Anísio Teixeira), accessible at <https://www.gov.br/inep/pt-br/aceso-a-informacao/dados-abertos/microdados>. It contains annual microdata from the National Basic Education Census spanning 2010 to 2020, with a separate CSV file representing each year.

The dataset carries the following characteristics:

- Format: Semicolon-delimited CSV files encoded in Latin-1, one file issued per census year (2010–2020)

- Columns: Each file holds approximately 370 attributes, capturing details on school infrastructure, staffing, student enrolment, geographic location, and institutional classification
- Volume: Roughly 300,000 records per year, amounting to around 3 million rows across the full eleven-year span
- Storage: Individual files average approximately 180 MB each, with the complete collection totalling around 2.2 GB
- Data types: A mix of integer codes (such as school type and region identifiers), string fields (municipality and state names), and numeric counts (enrollment and teacher figures).

2.2 Key Attributes Groups

Given the breadth of the dataset, the 370 available columns can be organised into four thematic categories:

- Geographic location: Includes state name (NO_UF), state abbreviation (SG_UF), state code (CO_UF), municipality name (NO_MUNICIPIO), and municipality code (CO_MUNICIPIO)
- School characteristics: Captures institutional type, administrative category, urban or rural classification, and dependency type
- Enrolment metrics: Records total student enrolment broken down by education stage early childhood, primary, and secondary as well as by gender and racial or ethnic group
- Infrastructure indicators: Flags the presence or absence of essential facilities including electricity, water supply, internet connectivity, a library, science laboratories, and sports areas

2.3 Relational Data Model

In its raw form, the source data is entirely flat and denormalised, with each row representing a single school's record for a given census year. All geographic, institutional, and enrolment information is embedded within that single row, resulting in considerable data redundancy. To

conceptually frame how this data is structured before any transformation takes place, the following logical entities and their relationships have been identified:

- SCHOOL (CO_ENTIDADE [PK], school name, institutional type, administrative category, location type) - uniquely identifies each educational institution
- LOCATION (CO_MUNICIPIO [PK], NO_MUNICIPIO, CO_UF, SG_UF, NO_UF) - represents the municipality and state to which each school belongs
- CENSUS_YEAR (NU_ANO_CENSO [PK]) - a year dimension that anchors each school record to its respective census period
- SCHOOL_RECORD (CO_ENTIDADE [FK], NU_ANO_CENSO [FK], CO_MUNICIPIO [FK], enrolment counts, teacher counts, infrastructure flags) - the central record linking a school, its location, and the census year

In terms of relationships, each SCHOOL_RECORD maintains a many-to-one association with SCHOOL, LOCATION, and CENSUS_YEAR respectively. Additionally, each SCHOOL is tied to a single LOCATION, reflecting the fact that every institution belongs to one municipality.

3.0 Data Preparation

This section describes the methodical process for converting unstructured, raw source data into an optimal, clean format that will be used for analytical modeling. The data preparation stage which focuses on reducing storage footprints, maintaining schema consistency and removing unnecessary data fills the gap between initial extraction and organized storage. Before entering the modeling pipeline, the data is refined to guarantee high performance and integrity through a series of sequential procedures that include automated ingestion and columnar file conversion.

3.1 Data Extraction

The first step in data preparation is to obtain the raw CSV files from the Brazilian government's Open Data portal using a Python script that launches Linux shell commands via the os module. The script downloads the relevant ZIP archives for each year between 2010 and 2020 which

extracts the CSV files and saves them in a local ./data directory. This stage uses the requests module for HTTP downloads and is entirely Python-based in which it requires no Spark.

3.2 CSV to Parquet Conversion

A SparkSession with the following configurations is used to read each CSV file: header=True, delimiter=";" and inferSchema=True. Spark only generates a logical reference to the file at this stage due to its slow evaluation which prevents any data from being loaded into memory. The Parquet format is then used to write each DataFrame to disk. With its columnar storage, built-in compression and predicate pushdown, Parquet produces files that are around five times smaller than their CSV equivalents and much faster for subsequent reads. All further processing reads from Parquet rather than the original CSV files after conversion.

3.3 Column Selection

The selection is limited to the amount of columns that are relevant to the intended Star Schema. This phase reduces memory usage by narrowing the dataset to only the information needed for the fact and dimension tables. Spark's distinct() function which eliminates null-key combinations, implicitly handles missing values in dimension columns, guarantees that dimension tables only contain valid, distinct entries.

3.4 Surrogate Key Generation

Every dimension table needs a primary key that is a distinct integer surrogate. After the data has been loaded, a follow-up SQL script is run using the psycopg2 library to apply the PRIMARY KEY constraint to each dimension table in PostgreSQL since Spark does not automatically enforce primary keys when it is published. This step procedure makes use of Spark's effective bulk write capability while guaranteeing referential integrity in the relational database.

4.0 Modeling

The data ecosystem's structural and architectural architecture is described in this section along with the methods used to analyze, arrange and use the produced data for business intelligence. It

describes the complete Apache Spark ETL pipeline that transfers data into a containerized relational environment from local storage. Moreover, this section covers the unified technological stack that underpins the complete analytical infrastructure and explains the specific dimensional architecture of the Star Schema including the exact schema definitions for the fact and dimension tables.

4.1 Architecture: Apache Spark ETL Pipeline

The pipeline architecture is divided into four stages: (1) Extract, which uses Python to download data from government portal; (2) Convert, which uses PySpark to convert CSV files to Parquet for performance; (3) Transform, which uses lazy evaluation to perform all dimensional modeling transformations in memory and (4) Load, which uses Java Database Connectivity (JDBC) to write the resulting DataFrames to a PostgreSQL instance running in Docker. In order to provide dashboards, Metabase is set up as a distinct Docker container that connects to PostgreSQL. Docker Compose is used to orchestrate all components on a single computer.

4.2 Multidimensional Data Model (Star Schema)

Star schema consists of only one Fact Table and many Dimension Tables. It is created with the purpose of making it easier to aggregate queries and provide information for Business Intelligence reports by eliminating the need to join tables to calculate aggregated values. Figure 4.1 displays the Censo Escolar DW database to showcase the single Fact Censo table alongside its corresponding dimension tables in a classic star schema design.

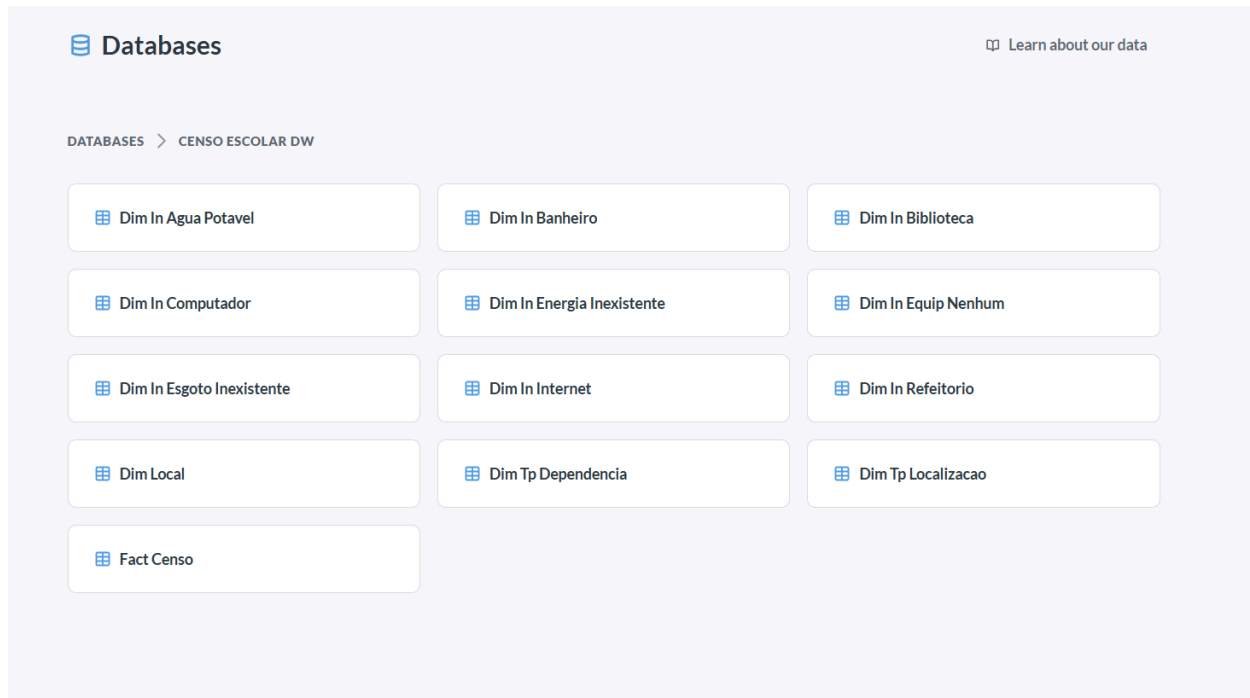


Figure 4.1: Databases

Dimension Tables include the following:

- Dim Local (id_location [PK], NO_UF, SG_UF, CO_UF, NO_MUNICIPIO, CO_MUNICIPIO) - geographic dimension capturing state and municipality details.
- Dim Tp Dependencia (id_school_dependency [PK], TP_DEPENDENCIA) - an operational dimension capturing school administrative and dependency data.
- Dim Tp Localizacao (id_school_location [PK], TP_LOCALIZACAO) - an operational dimension capturing urban or rural school classifications.

Binary Infrastructure Dimensions is to maximize query performance inside Metabase, individual infrastructure characteristics have been isolated into granular, specific lookup tables :

- Dim In Agua Potavel (id_agua [PK], binary flag for drinking water availability)
- Dim In Banheiro (id_banheiro [PK], binary flag for toilet facilities)
- Dim In Biblioteca (id_biblioteca [PK], binary flag for library presence)
- Dim In Computador (id_computador [PK], binary flag for computer availability)
- Dim In Energia Inexistente (id_energia [PK], binary flag for lack of electricity)
- Dim In Esgoto Inexistente (id_esgoto [PK], binary flag for lack of sewage systems)
- Dim In Internet (id_internet [PK], binary flag for internet connectivity)

- Dim In Refeitorio (id_refeitorio [PK], binary flag for cafeteria presence)
- Dim In Equip Nenhum (id_equip [PK], binary flag for no school equipment)

The Fact Table looks as follows:

- fact_enrollment (id_fact [PK], id_location [FK], id_school_type [FK], id_race [FK], id_year [FK], QT_MAT_INF, QT_MAT_FUND, QT_MAT_MED, QT_DOC_BAS, QT_SALAS_EXISTENTES)

The measures are stored in the fact table, where the dimensions have foreign keys linked to them. The Fact Table contains the metric values (count of enrolled students for each level of education, teacher's count, class rooms' count), with one row representing a school per year census report. The Fact Table can be loaded using the append technique to maintain historical data from every year. It facilitates quick GROUP BY queries on the combination of dimensions used in Metabase.

4.3 Technology Stack

To implement the proposed multidimensional data model and ensure efficient data processing, storage and visualization, a modern-open source data engineering stack is utilized. The specific technologies and tools selected for the ETL pipeline and Business Intelligence layer include:

- Apache Spark (PySpark): Distributed data processing engine for ETL transformations
- PostgreSQL (via Docker): Relational database for storing the Star Schema
- Metabase (via Docker): BI dashboard tool for visualising query results
- Python 3.6+ with psycopg2 (PostgreSQL connection) and requests (HTTP download)
- Parquet: Columnar file format used as intermediate storage between CSV and the database load.

5.0 Evaluation

This section provides a comprehensive assessment of the developed data engineering solutions, evaluating its technical performance, data reliability and strategic value. By analyzing the

pipeline's efficiency, data integrity and alignment with initial business objectives, we can measure the success of the architecture and identify critical areas for future optimization.

5.1 Pipeline Performance

The entire process involved in loading, transformation and ingestion of all eleven years of census data (~3M rows) to the star schema was successful in an acceptable runtime. By converting the dataset from CSV format to Parquet, there is about 80% savings of the space required to store the dataset and improved read performance due to columnar storage and predicate pushdown capabilities. Creation of dimension tables takes less than two minutes. Loading of fact tables is the most resource-intensive part as it requires joining the data in the source to all dimension tables to generate surrogate keys.

5.2 Schema Correctness and Data Integrity

Star schema generated in this way is validated via execution of queries to PostgreSQL database, in which uniqueness of data is confirmed in dimension tables. For example, 27 unique states in the dim_location table, 11 unique years in the dim_year table. The row count of the fact table is consistent with the expected number of approximately 3 million rows. Integrity constraint is guaranteed since each foreign key of the fact_enrollment table corresponds to the primary key from the appropriate dimension table, validated through psycopg2 PRIMARY KEY and FOREIGN KEY constraints.

5.3 Business Objective Alignment

The Metabase dashboards demonstrate that the Star Schema successfully supports the original business questions. Example dashboard panels include total enrollment by year (a time-series trend chart), total enrollments by state in 2021 (a choropleth or bar chart by state) and the top 10 cities by enrollment year. The creation of dashboards proves that the data pipeline has succeeded in meeting the business goal of allowing quick and flexible analysis queries for educational data. Queries that were previously taking time because they involved importing the 2.2 GB CSV file using Pandas can now be done in seconds.

To visually demonstrate these capabilities, key panels from the Metabase instances are presented below as in Figure 5.1 and 5.2:

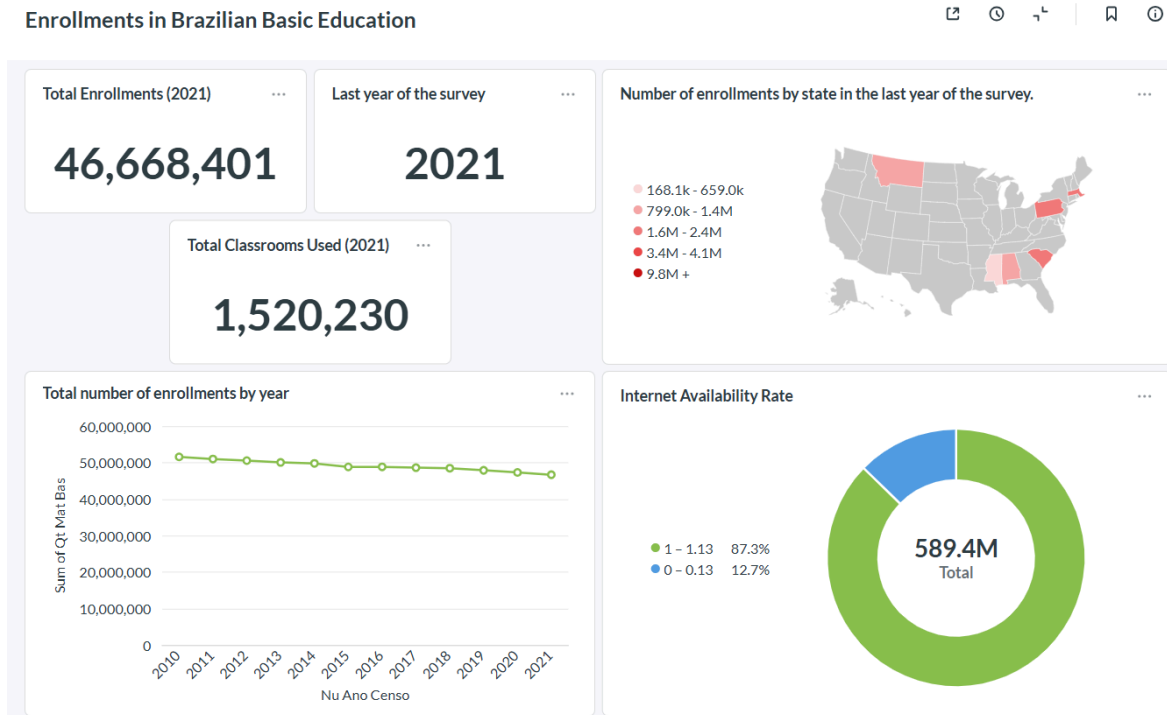


Figure 5.1: Dashboard Overview (Upper Section)

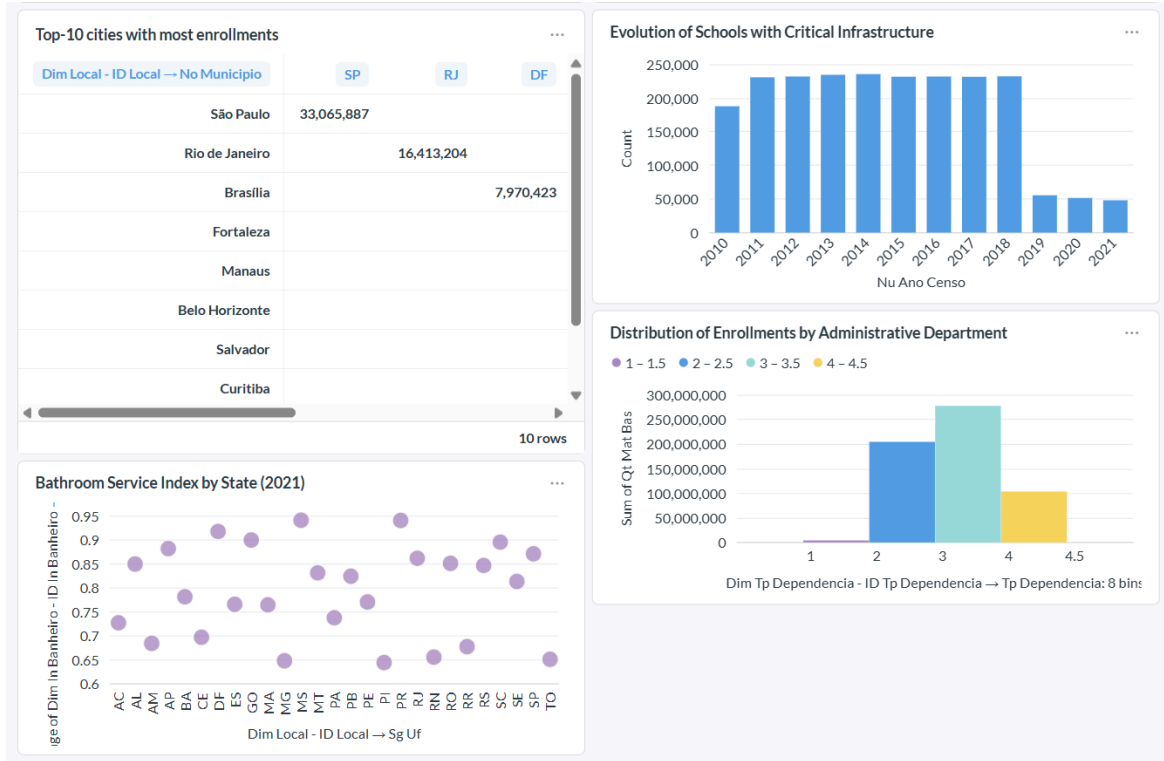


Figure 5.2: Dashboard Overview (Lower Section)

5.4 Limitations and Potential Improvements

While the current implementation successfully establishes the end-to-end pipeline, certain constraints exist regarding scalability, data governance and workflow automation. The following points highlight the system's current limitations alongside recommended advancements for future iterations:

- The pipeline operates within the boundaries of a single local computer. A genuine distributed Spark cluster is necessary for actual big-data projects.
- The problem of missing values in non-key attributes as well as problems with consistent coding in the yearly years cannot be addressed directly. Schema inference is all that can be done about it.
- No automation for scheduling occurs. In order to run the pipeline incrementally for every new year of census data, an orchestration tool like Apache Airflow needs to be used.
- Portuguese-to-English column mappings are rough approximations. A standardized mapping dictionary will make the dashboard easier to read internationally.

6.0 Reflection

Safiya Nursyahadah binti Masnoor

Working on this project was both challenging and rewarding. One of the biggest difficulties I faced was dealing with the sheer size of the dataset processing over 2.2 GB of CSV files was something I had never done before and getting Apache Spark to work correctly took a lot of trial and error. Redesigning the flat data into a Star Schema was also tricky at first, as it required me to think carefully about how the data related to one another. However, working through these challenges helped me gain a much better understanding of how real-world data pipelines are built and why proper data modelling matters. By the end of the project, I felt more confident in my ability to handle large datasets and apply tools like PySpark and PostgreSQL in a meaningful way.

Farra Nurzahin binti Zaharil Anuar

This tutorial helped me understand how an Apache Spark ETL pipeline works from start to finish. I learned how to identify business problems, understand large datasets and transform data into a better structure. I also gained experience in extracting CSV files and converting them into Parquet format. In addition, I learned how Star Schema, Apache Spark, PostgreSQL, Docker, and Metabase work together in one pipeline. Overall, this tutorial improved my understanding of big data processing and increased my confidence in using Apache Spark for data engineering tasks.

Dayang Farah Farzana binti Abang Idham

Participating in this tutorial has been a challenging journey that drastically evolved my practical capabilities when managing a 2.2 GB Brazilian Education census dataset. One of the most complex technical challenges I faced was configuring the containerized infrastructure via Docker Compose and troubleshooting localized Command Prompt connection bottlenecks where the execution environment halted due to inactive Docker states. To overcome these challenges, we established a strong team dynamic where all group members actively searched for solutions on the Internet and successfully troubleshoot the problems and fix them. Through this project, I gained valuable skills in container orchestration, network troubleshooting and star schema modelling. This experience has given me the technical knowledge to build scalable big data solutions in the future.